KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Computer Science and Engineering

Spring-2022 Final Examination

Program: B.Sc. in CSE (Batch – 12^{th}) 1^{st} Year 2^{nd} Semester
Course Title: Physics - II
Course Code: PHY 1201

PART-A: MCQ

Time: 10 Minutes **Total Marks: 10x1=10**

Student's ID:																

Date:									**Invigilator's Signature:**	

(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

1. Which is essential for interference?
   A) Two different color lights.
   B) The movement of the two waves should be opposite.
   C) The light sources should be far apart.
   D) The light sources should be coherent.
   E) The phase difference should have to be $\frac{\pi}{2}$.

2. Corpuscular theory of light given by _______ .
   A) Newton
   B) Maxwell
   C) Max Plank
   D) Huygen
   E) Fermat

3. Speed of the electromagnetic wave depends on ________ of the medium.
   A) elasticity
   B) molecular structure
   C) radioactivity
   D) density
   E) magnetic permeability

4. The phase difference between two points on a wave is π, what is the path difference between them?
   A) λ
   B) $\frac{\lambda}{2}$
   C) $\frac{\lambda}{4}$
   D) $\frac{3\lambda}{2}$
   E) $\frac{3\lambda}{4}$

5. What is the Brewster angle for air to glass transition? (Refractive index of glass = 1.5)
   A) 36.31^0
   B) 46.31^0
   C) 56.31^0
   D) 66.31^0
   E) 76.31^0

6. What will be the angle between the reflected ray and refracted ray when the light ray incident at a polarizing angle to a surface?

A) 90^0

B) 60^0

C) 45^0

D) 30^0

E) 0^0

7. The radius of the hydrogen nucleus is ________________ .

A) 0.53 m

B) 1.2×10^{-15} m

C) 3.2×10^{-15} m

D) 13.6×10^{-15} m

E) 17.6×10^{-15} m

8. Half-life of a radioactive substance is 10 days. How long will it take to decrease to 25% of it?

A) 5 days

B) 10 days

C) 15 days

D) 20 days

E) 25 days

9. What is the unit of the decay constant?

A) m

B) kg

C) J

D) Sec

E) A

10. A space shuttle crosses the football ground of KYAU along of its length at speed of $0.6c$. The football ground is $100m$ long and $50m$ wide. What will be width of the football ground according to the measurement by a foreign astronaut in the space shuttle?

A) $100m$

B) $80m$

C) $50m$

D) $40m$

E) $30m$

Signature of Examiner

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Course Title: Physics - II
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PART-B: SAQ

Time: 1 Hour and 50 Minutes **Total Marks: 5x6=30**

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. See the stem and answer the questions that follows:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow ({}^{236}_{92}\mathrm{U})^* \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} + \mathrm{Energy}$$

Where, $M({}^{235}_{92}U) = 235.0439\, amu$; $M({}^{141}_{56}Ba) = 140.9139\, amu$;

$M({}^{92}_{36}Kr) = 91.8973\, amu$; $M({}^{1}_{0}n) = 1.0087\, amu$

a) Give a list of the main components of the nuclear reactor. 1

b) Calculate the value of the energy of above stem. 2

c) Interpret the process of expansion of the above reaction in the reactor and specify a controlling mechanism for that expansion. 3

2. a) Define stopping potential. 1

b) How are photoelectrons generated? 2

c) UV light of wavelength 350 nm and intensity 1.00 w/m^2 is directed at a potassium surface. Find the maximum K.E. of photoelectrons. (Work function of Potassium is 2.2 eV) 3

3 a) Distinguish the inertial and non-inertial frame of references. 2

b) Is it possible that the mass of an electron is equal to the mass of a proton? Analyze mathematically. 4

4. a) Define wave front. 1

b) Illustrate the light interference that occurs in a thin film. 2

c) A thin sheet of Mica (μ = 1.58) is used to cover one slit of the double-slit arrangement. The central point on the screen is now occupied by what to be the seventh bright fringe. If the wavelength of the light used is $5500 A.U$. What is the thickness of the Mica sheet? 3

5. a) What is the dispersive power of grating? 1

b) Draw the intensity distribution curve for diffraction of light. 2

c) How many orders will be visible (Max. observable angle is 30^0) if the wavelength of incident radiation be $5000 A.U$. and the number of lines on the grating be 14000 per inch? 3

6. a) State Malus' Law. 1

b) Write some methods of polarization of light. 2

c) If the refractive index of a polarizer is 1.9218. What will be the polarization angle and angle of refraction? 3